

Assignment 2: Learning from Click-Logs on EB-NeRD and MIND

CS4.406: Information Retrieval & Extraction
News Recommendation Systems

Due: September 20, 2026

Type	Teams of 2
Prerequisite	Assignment 1 (Lexical & Semantic Retrieval)
Report Submission	Moodle
Competitions	MIND (Codabench) RecSys 2024 Challenge (Codabench)
Starter Code (EB-NeRD)	https://github.com/jppol-ai/ebnerd-benchmark
MIND Dataset	https://huggingface.co/datasets/yjw1029/MIND
EB-NeRD Dataset	https://recsys.eb.dk/dataset/

***Note:** You must register on Codabench and submit to both leaderboards.*

Introduction

Build on the lexical and semantic retrieval pipeline from Assignment 1 to add **behavioural signals from click-logs**. You will implement a re-ranker, reproduce a baseline then beat it with an ablation, and analyse serving and scale. You will continue working with both MIND and EB-NeRD datasets. This is a **team assignment** — work in groups of two.

Important

Mandatory: You **must** register for both competitions on Codabench and submit your predictions to both leaderboards. Topping a leaderboard is not a necessity—though always appreciated—and grading is **never** on leaderboard rank. Register here:

- **MIND Competition:** <https://www.codabench.org/competitions/13967/>
- **RecSys 2024 Challenge:** <https://www.codabench.org/competitions/2469/>

Datasets

EB-NeRD (RecSys 2024 Challenge, Ekstra Bladet) [Kruse et al., 2024]: ~2.7M users, 600M+ impression logs, 120K+ articles with title/abstract/body and provided article embeddings. Demo/small/large bundles let you dial scale gradually. Danish language. **The large bundle (ebnerd_large.zip + ebnerd_testset.zip) is required for Codabench submission** — the test set has 13.5M impressions with no click labels.

MIND (Microsoft News Dataset) [Wu et al., 2020]: ~1M users, 160K+ English news articles with title/abstract/body and entity annotations, 15M+ impression logs. MIND-small for fast

iteration. **The large test set (MINDlarge_test.zip) is required for Codabench submission** — it contains 2.37M impressions with no click labels.

Both tasks natively exercise all three modelling axes:

- **Lexical** — titles/bodies (BM25, TF-IDF)
- **Semantic** — article/text embeddings (Word2Vec, BERT, XLM-RoBERTa)
- **Behavioural** — click history, recency/decay, session context

Rapid news decay makes temporal splitting and freshness handling instructive. Both leaderboards are live, so you can submit and verify your solutions against them.

What You Will Build

1. **Two-stage retrieve-then-rank** — use Assignment 1’s candidate generator to retrieve top- K candidates, then train a re-ranker (GBDT or a small neural ranker) over engineered behavioural features.
2. **Click-history & session features** — recency-weighted history, session context, article popularity/freshness, with strict behaviour-window boundary enforcement.
3. **Baseline reproduced, then beaten** — reproduce the official/starter baseline, then one principled improvement with an ablation; claimed gains must ship a paired bootstrap 95% CI that excludes zero.
4. **Serving & scale analysis** — index memory, p99 retrieval latency, and a back-of-envelope cost/QPS at a target SLA.
5. **Extended evaluation** — all metrics (AUC, MRR, nDCG@5, nDCG@10, diversity, novelty, coverage) with slicing and bootstrap CIs.
6. **Report** — what you built, choices and alternatives, observations, and where it breaks at $10\times$.

Part 0: Datasets & Setup

This assignment builds on Assignment 1. Download the datasets if you haven’t already.

Important

Large datasets are required for Codabench submissions. The small/demo sets are useful for development and debugging, but the test sets used by both leaderboards come from the **large** bundles only. Make sure to download the large test sets early — they are several GB each.

EB-NeRD — download from S3:

```
1 # Small/demo for development
2 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/ebnerd_demo.zip
3 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/ebnerd_small.zip
4
5 # LARGE set (required for Codabench submission)
6 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/ebnerd_large.zip
```

```

7 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/artifacts/articles_large_only.zip
8 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/ebnerd_testset.zip
9
10 # Pre-computed embeddings
11 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/artifacts/Ekstra_Bladet_word2vec.zip
12 wget https://ebnerd-dataset.s3.eu-west-1.amazonaws.com/artifacts/
    google_bert_base_multilingual_cased.zip

```

MIND — download from HuggingFace:

```

1 # Small set for development
2 hf download yjw1029/MIND --repo-type dataset --local-dir MIND_data
3
4 # Or individual files:
5 # MINDsmall_train.zip, MINDsmall_dev.zip -- for development
6 # MINDlarge_test.zip -- required for Codabench submission
7 # MINDlarge_train.zip, MINDlarge_dev.zip -- full-scale training

```

Competition Registration: Register on both leaderboards:

- **MIND:** <https://www.codabench.org/competitions/13967/>
- **RecSys 2024 Challenge:** <https://www.codabench.org/competitions/2469/>

Compute: Use free-tier GPUs (Google Colab, Kaggle, Lightning AI). The large EB-NeRD test set has 13.5M impressions and the MIND large test has 2.37M impressions — ensure your prediction pipeline is memory-efficient (use Polars, PyArrow, or batch processing).

Part I: Learning from Click-Logs

Team assignment (groups of 2). Due: September 20, 2026.

Building on Assignment 1, you will add behavioural signals from click-logs, implement a re-ranker, reproduce a baseline then beat it with an ablation, and analyse serving and scale.

Q1. Click-History & Session Features

Engineer behavioural features from the click-logs for both datasets:

1. **Click-history features:** User's recent clicked articles (titles, categories, embeddings), click count, recency-weighted history (exponential decay)
2. **Session features:** Within-session click patterns, dwell time (if available), position bias
3. **Article features:** Popularity, freshness (time since publish), category match with user history
4. **Behavioural-window boundary:** Enforce that no future clicks leak into features at training or serving time

Q2. Re-Ranker

Implement a two-stage retrieve-then-rank pipeline:

1. Use Assignment 1's candidate generator to retrieve top- K candidates ($K \sim 100\text{--}200$)

2. Train a re-ranker over the engineered behavioural features:
 - Option A: GBDT (LightGBM/XGBoost) over hand-crafted features
 - Option B: A small neural ranker (e.g., NRMS-style [Wu et al., 2019] or a simple MLP)
3. Score and re-rank the candidates
4. Report AUC, MRR, nDCG@5, nDCG@10 before and after re-ranking

Q3. Baseline Reproduced, Then Beaten

1. **Reproduce** the official/starter baseline (e.g., NRMS from the `ebnerd-benchmark` repo, or the MIND baseline) on both datasets
2. **Improve** with one principled change (e.g., better user encoding, attention over history, category-aware features, freshness weighting)
3. **Ablation:** Run an ablation study isolating the contribution of your improvement
4. **Statistical significance:** Claimed gains must ship a **paired bootstrap 95% CI** that excludes zero

Q4. Serving & Scale Analysis

Analyse the serving characteristics of your system:

1. **Index memory:** Measure the memory footprint of your ANN index and feature store
2. **Latency:** Measure p99 retrieval latency (candidate generation + re-ranking) for a single user request
3. **Cost/QPS:** Back-of-envelope estimate of cost per 1000 queries at a target SLA (e.g., p99 < 100ms)
4. **Scaling argument:** How would your system scale to 10× the current load? What breaks first?
5. A measured local benchmark plus a scaling argument suffices

Q5. Extended Evaluation

Extend the evaluation harness from Assignment 1:

- Report all metrics (AUC, MRR, nDCG@5, nDCG@10, diversity, novelty, coverage) for the full two-stage pipeline
- Include at least two slices: cold-start vs. warm, head vs. tail
- Bootstrap 95% CI for all reported metrics
- Submit final predictions to both Codabench leaderboards and include screenshots

Q6. Design Note

Write a concise **design note as a PDF** (6 pages target, 11pt, 1-inch margins) covering:

- What you built and key design choices (re-ranker architecture, features)
- Baseline vs. improved results with ablation and CI
- Serving and scale analysis findings
- Where your system breaks at $10\times$ scale

The 6-page target is a guideline, not a hard cap — you may extend it if your content justifies the extra length, but avoid padding. References and appendices do not count toward the target.

Part II: Deliverables & Policies

Q7. Deliverables

1. **Code**: reproducible pipeline, model code, evaluation harness, prediction files, `README.md` with one-command reproduce. No large files — use `.gitignore`.
2. **Report** (Moodle): what you built, choices, observations, where it breaks at $10\times$.
3. **Leaderboard screenshots** from both Codabench competitions.
4. **AI usage log**: all prompts, chat history exports, marking of AI-generated vs. human-written code.

Q8. Git Commit Policy

- Commit frequently with meaningful messages.
- No large files in Git — ignore `*.zip`, `*.pt`, `*.ckpt`, `__pycache__`/, `data/`.
- No force-pushes after the deadline.

Q9. Anti-Gaming

- Report metrics with and without features unavailable at serving time.
- Enforce the behaviour-window boundary — no future-click leakage. Include a test asserting this.

Important

Grading note: Grading is **never** on leaderboard rank. Your grade is based on pipeline correctness, system design, ablation rigour, scale analysis, and design-note clarity.

References

Johannes Kruse, Kasper Lindschow, Saikishore Kalloori, Marco Polignano, Claudio Pomo, Abhishek Srivastava, Anshuk Uppal, Michael Riis Andersen, and Jes Frellsen. Eb-nerd: A large-scale dataset for news recommendation. In *Proceedings of the ACM RecSys Challenge 2024*, 2024.

Chuhan Wu, Fangzhao Wu, Suyu Ge, Tao Qi, Yongfeng Huang, and Xing Xie. Neural news recommendation with multi-head self-attention. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6389–6394, 2019.

Fangzhao Wu, Ying Qiao, Jiun-Hung Chen, Chuhan Wu, Tao Qi, Jianxun Lian, Danyang Liu, Xing Xie, Jianfeng Gao, Winnie Wu, and Ming Zhou. Mind: A large-scale dataset for news recommendation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3597–3606, 2020.